RESEARCH PRESENTATION

MERGE: Multimodal Emotion-Driven Rhythmic Lyric Generation Engine

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ABSTRACT

1-Slide Summary

- **Problem:** Existing systems focus on either emotion OR structure of lyrics, rarely both in a multimodal context.
- **Solution:** Proposed MERGE, a framework for rhythm-aware lyrics conditioned on facial expressions.
- **Method:** Infers emotional state via FER (Facial Emotion Recognition) to guide GPT-2 generation.
- **Evaluation:** Introduced **PHREM**, a comprehensive research-level metric for quality assessment.

PROBLEM STATEMENT

Structural Focus

Prior systems primarily emphasize rhyme at the cost of emotional depth.

Emotional Control

Limited research on using external modalities (visual/face) for linguistic creative steering.

Integration Gap

Absence of frameworks combining facial emotion vectors with rhythm-constrained text decoding.

Goal: Generate emotion-aware and rhythm-consistent lyrics through a unified multimodal pipeline.

MOTIVATION

Why Lyrics?

- **Complexity:** Direct music synthesis faces data scarcity and high audio dimensionality.
- **NLP Scope:** Lyric generation allows sophisticated semantic and structural study within text domains.
- **Expressivity:** Lyrics serve as the primary emotional carrier in modern musical compositions.

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RESEARCH QUESTIONS

RQ1: Control Signal Efficacy

Can facial emotion signals effectively serve as control inputs for fine-grained lyric generation?

RQ2: Constraint Interaction

Can rhythmic constraints be enforced during decoding without significantly degrading emotional alignment or linguistic fluency?

CORE CONTRIBUTIONS

MERGE Framework

First end-to-end framework connecting FER with rhythmic lyric generation.

Annotation Pipeline

Scalable method to map GoEmotions to FER classes for large-scale labeling.

PHREM Metric

Advanced evaluation metric capturing phonetic flow, rhythm, and coherence.

I. Background & Literature

Current state of Facial Emotion Recognition and Controlled Text Generation

RELATED WORK: FER

Facial Emotion Recognition

- Evolution from ResNet/MobileNet to Transformer-based architectures.
- Benchmark datasets: FER-2013 (35k images) and RAF-DB.
- Focus on 7-class classification (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral).

RELATED WORK: TEXT GEN

Controlled Text Generation

- Attribute-guided generation using control tokens (Radford et al., 2019).
- Poetry systems using CMU Dictionary for phonetic rules (Ghazvininejad et al., 2016).
- **Missing:** Systems that merge multimodal visual cues with multi-attribute text steering.

II. Dataset Details

Large-scale lyric corpora and multi-dataset FER unification

DATA SOURCES

Lyrics: Genius Corpus

- ~3.2 Million English-language songs.
- Multi-genre diversity for robust training.
- Filtered for high-quality metadata.

Vision: FER Unified

- **FER-2013:** Grayscale baseline.
- **RAF-DB:** Real-world images.
- **AffectNet:** High-scale annotation (30k used).

EMOTION TAXONOMY

To ensure cross-modal compatibility, we utilized a **6-class mapping**:

Angry	**Fear**	**Happy**
Sad	**Surprise**	**Neutral**

Exclusion of Disgust: Removed due to severe class imbalance in FER datasets and significant linguistic overlap with 'Sadness' in lyrics.

EMOTION MAPPING PIPELINE

GoEmotions to FER

- Raw lyrics lack explicit emotion labels.
- **Step 1:** 28 GoEmotions categories predicted line-by-line.
- **Step 2:** Line probabilities grouped into 6 parent FER sets.
- **Step 3:** Weighted aggregation for song-level sentiment.

MATHEMATICAL FORMULATION

Probability vector aggregation for a song with n lines:

$$Y_{song} = \frac{\sum_{i=1}^{n} w_i q_i}{\sum_{i=1}^{n} w_i}$$

Where $w_i = 1 - q_{neutral}$ represents emotional intensity. The dominant emotion e^* is selected via $\arg\max$.

III. Methodology

The MERGE System Architecture & Constrained Decoding

SYSTEM OVERVIEW

The Pipeline

1. **Input:** User Face Image.
2. **FER Module:** Predicts emotion (e.g., Happy).
3. **Control Token:** Prepends to GPT-2.
4. **Transformer:** Generates next-token probability.
5. **Constraint Decoder:** Filters tokens for rhythm/rhyme.

FACIAL EMOTION RECOGNITION

Model Specs

- CNN-based Feature Extractor.
- Softmax layer for probability output across 6 classes.
- Progressive unfreezing to prevent overfitting on FER-2013 benchmarks.

Training Strategy: Phase V utilized an ensemble (EfficientNet-B3 + DenseNet-121) for superior robustness across multi-dataset variance.

EMOTION CONDITIONING

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[User Image] → FER →  → GPT-2 → "Sunshine in my pocket..."
```

Emotion control is implemented via explicit control tokens. The model learns associations between high-level tags and thematic clusters (e.g., Happy → Imagery of celebration/light).

LANGUAGE MODEL: GPT-2

Lyric generation is a conditional probability task:

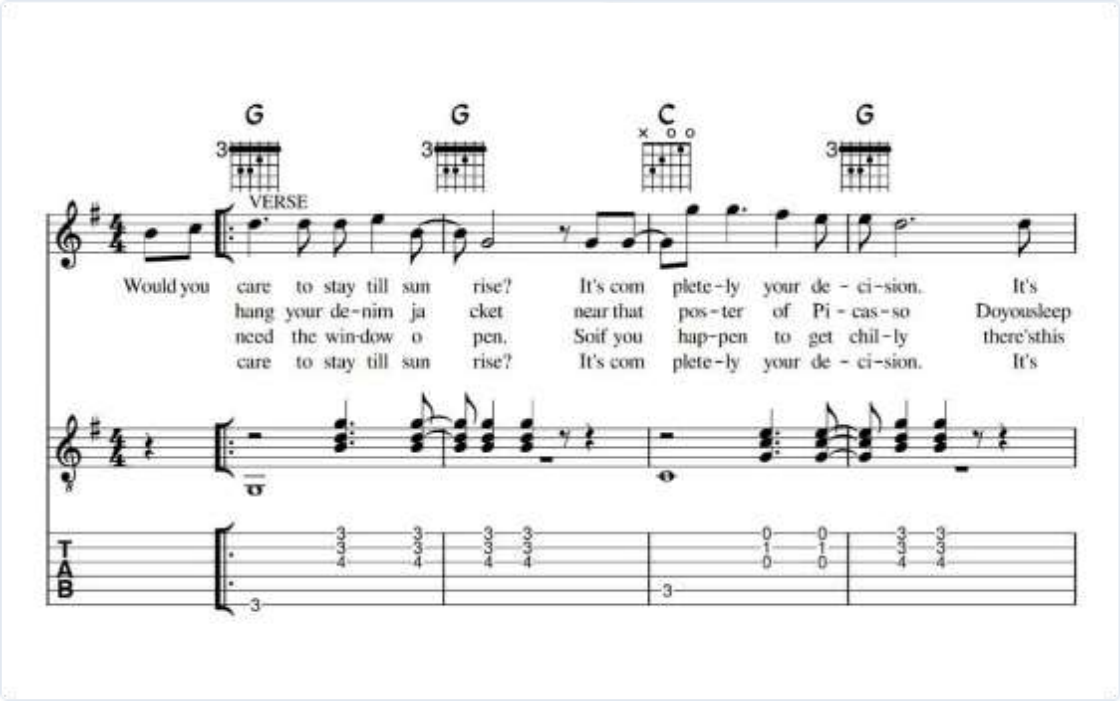
$$P(x \mid e) = \prod_{t=1}^{T} P(x_t \mid x_{<t}, e)$$

- **Optimizer:** AdamW with linear warmup.
- **Sequence Length:** 512 tokens.
- **Finetuning:** Guided by mapped emotional Genius dataset.

RHYTHM-CONSTRAINED DECODING

Constraint Enforcement

- **Syllable Count:** Target counts maintained via candidate filtering.
- **Rhyme Schemes:** CMU Dictionary lookup for AABB/ABAB patterns.
- **Soft Back-off:** Strategy allows line regeneration if decoding becomes infeasible.



IV. The PHREM Metric

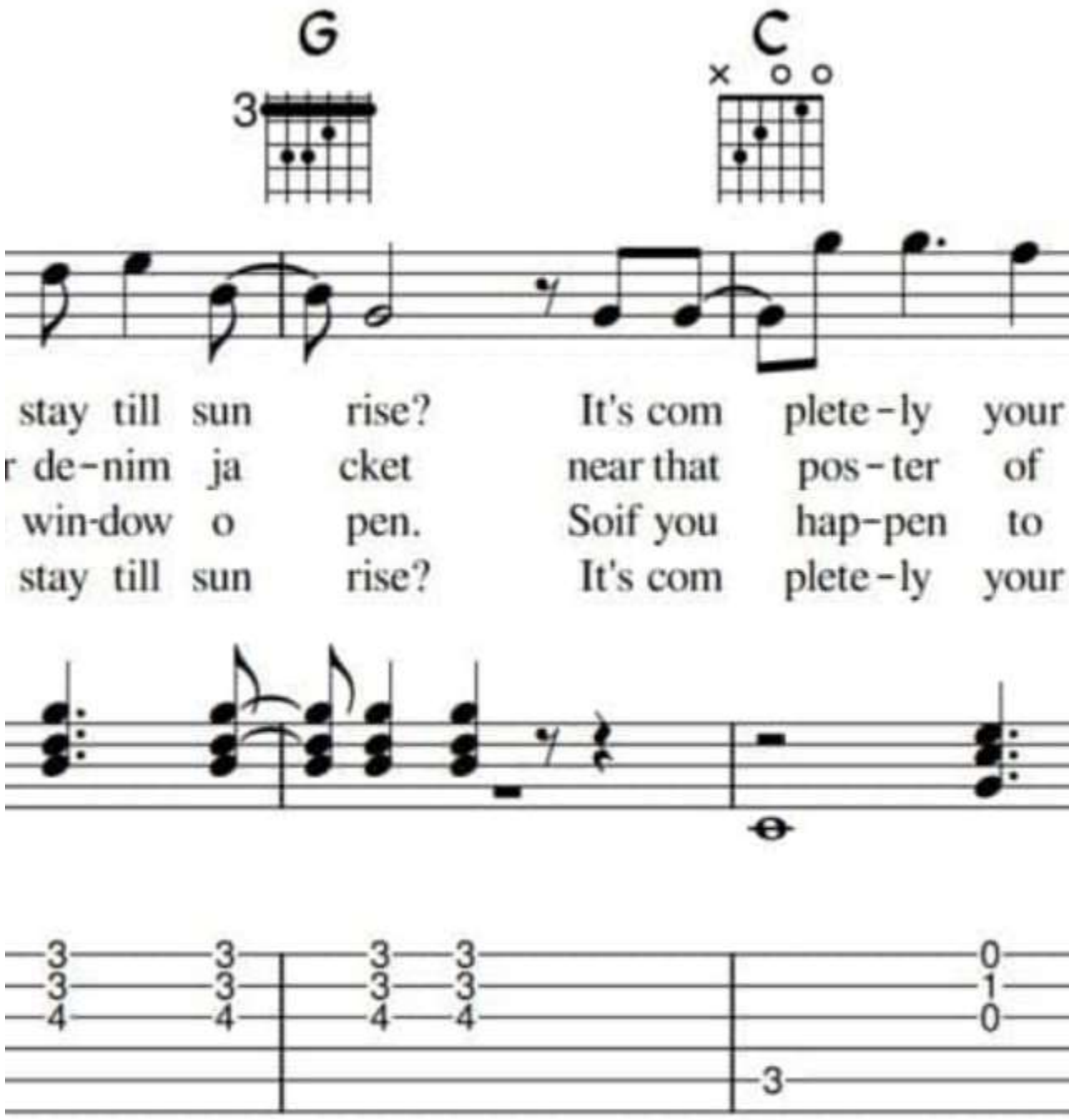
Phonetic Rhythm Emotion Metric: A Lyrical Evaluation Standard

WHY PHREM?

Existing NLP metrics like **BLEU** or **ROUGE** fail for lyrics because they prioritize lexical overlap over musicality.

Lyrics require a holistic view of:

- Phonetic flow
- Rhythm/Meter
- Emotional Arc
- Semantic Coherence



PHREM CALCULATION

PHREM is a composite metric computed as a weighted sum of 7 dimensions:

$$\text{PHREM} = \sum_{i=1}^{7} \lambda_i C_i$$

Priority weights: **EAS (0.22)** for emotion and **MCS (0.20)** for coherence ensure thematic alignment remains primary.

FLOW & RHYTHM

PPFS: Phonetic Flow

Smoothness of phoneme transitions. Penalizes jarring vowel-consonant shifts.

RSFS: Rhythmic Structure

Measures syllable uniformity and stress pattern regularity across lines.

EMOTION & CATCHINESS

EAS: Emotion Arc Score

Measures emotional continuity across line transitions using JS divergence.

HQCS: Hook Quality

Identifies repeated n-gram patterns and rewards catchy refrains.

RHYME & COHERENCE

RCS: Rhyme Consistency

Evaluates rhyme quality of line-final words via phonetic pattern matching.

MCS: Motif Consistency

Topical coherence measured via sentence embeddings (Local vs. Global similarity).

DIVERSITY & LENGTH

DPS: Degeneracy Penalty

Exponential penalty for excessive lexical repetition (word loop detection).

LPS: Length Penalty

Discourages very short lyrics (sub-8 lines) to ensure narrative substance.

V. Experiments & Results

FER Training Phases and GPT-2 Performance Analysis

FER TRAINING PHASES

- **Phases I-II:** Established ResNet18/34 baselines (Overfitting found).
- **Phase III:** ResNet50 with linear warmup and label smoothing.
- **Phase IV:** Reformulated to 6-class (Test Acc 67.43%).
- **Phase V:** Multi-dataset ensemble with EfficientNet-B3 backbone.

GPT-2 CONFIGURATION

- Prepended control tokens: e.g., , .
- Optimizer: AdamW (lr = 5e-5).
- Dataset split: 90% training / 10% validation for each emotion class.
- Metrics used during training: Cross-entropy loss and Perplexity.

FER PERFORMANCE RESULTS

Phase	Backbone	Test Accuracy	Train-Test Gap
Phase I	ResNet18	0.620	8-12%
Phase IV	ResNet50 (6-cls)	0.674	<2%
Phase V	Ensemble + TTA	0.751	~2%

The Phase V Ensemble showed significant generalization across unified datasets.

LYRIC GENERATION SAMPLES

Happy Condition

"Sunlight dancing on the floor,
Open up every single door,
Laughter rising in the air,
Joy is everywhere."

Sad Condition

"Empty hallways, shadows cold,
Stories left as yet untold,
Rain against the window pane,
Memories are all that remain."

ANALYSIS & DISCUSSION

Strengths

- Successful emotion conditioning.
- Scalable pipeline for million-scale song labeling.
- Reduced overfitting via progressive unfreezing.

Weaknesses

- **Fear class:** Consistent lowest F1 (0.448) due to subtle facial movements.
- **Rhythm:** Not fully enforced in all complex meter types.

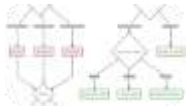
CONCLUSION & FUTURE WORK

MERGE effectively integrates **Visual Emotion** with **Rhythmic Language Modeling**. PHREM provides a robust quantitative baseline for creative lyric evaluation.

Future Horizons

- Deeper Rhythm Enforcement: Improving multi-beat patterns.
- Larger Models: Moving to GPT-3/LLaMA architectures.
- Human-in-the-loop: Incorporating subjective evaluation to calibrate PHREM weights.

IMAGE SOURCES



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